



Depression

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Introduction

In the *Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition, Text Revision* (DSM-5-TR), depressive disorders include^[1]:

- Major depressive disorder (MDD)
- Persistent depressive disorder
- Disruptive mood dysregulation disorder
- Premenstrual dysphoric disorder
- Substance-induced mood disorder (caused by substances such as alcohol or medications)
- Mood disorder due to a general medical condition (when the depression is thought to be a direct physiological consequence of the medical disorder)
- Other specified depressive disorder (depressive episodes deviating from the precise criteria for MDD)
- Unspecified depressive disorder

This chapter primarily focuses on major depressive disorder and persistent depressive disorder.

Major depressive episodes may be single or, more often, recurrent; see [Table 1](#) for diagnostic criteria. Persistent depressive disorder, or dysthymia, is a consolidation of dysthymic disorder and chronic major depressive disorder. In contrast to MDD, the symptoms of persistent depressive disorder are typically fewer and less severe. The diagnostic criteria for persistent depressive disorder include depressed mood for at least 2 years *and* at least 2 of the following: increased or decreased appetite, insomnia or hypersomnia, reduced energy or fatigue, reduced self-esteem, reduced concentration or ability to make decisions, and hopelessness.^[1]

The lifetime prevalence of MDD in Canada is approximately 12% and the annual prevalence of a major depressive episode is just under 5%, with women having a generally higher prevalence than men.

Depression has a negative impact on work productivity and is also associated with serious impairment in quality of life and a high economic burden.^[2]

Goals of Therapy

- Achieve remission of depressive symptoms
- Treat concomitant symptoms/disorders
- Prevent suicide
- Restore optimal functioning
- Prevent recurrence

Investigations

The following screening questions/assessment tools are recommended for use by primary health-care professionals:

- The [Patient Health Questionnaire \(PHQ-9\)](#)^[3] is a patient-rated assessment tool that consists of 9 questions that correspond to the DSM-5 criteria for a major depressive episode.^[4] The first 2 questions from the PHQ-9 serve as a useful and rapid screening tool:^[5]
 - During the past 2 weeks, how often have you been bothered by little interest or pleasure in doing things?
 - During the past 2 weeks, how often have you been bothered by feeling down, depressed or hopeless?
- The [Quick Inventory of Depressive Symptomatology-Self Report \(QIDS-SR16\)](#) is a widely used alternative to the PHQ-9. It is available in multiple languages, takes less than 10 minutes to complete and consists of 16 multiple choice questions that cover DSM-5 criteria.
- The 7-item [Hamilton Depression Rating Scale \(HAMD-7\)](#)^[6] is a validated, brief assessment for health-care professionals designed to rate severity and remission.
- The [Mood Disorder Questionnaire \(MDQ\)](#)^[7] is a useful screening instrument for manic or hypomanic symptoms.
- The [Edinburgh Postnatal Depression Scale \(EPDS\)](#)^[8] is the most widely used and well-validated tool to screen for depressive symptoms during pregnancy and the postpartum period.
- [MoodEx](#) is a free website that offers tools for patients to use in measurement-based care—the use of measurement tools to assess disease progress and aid in therapeutic decisions.

As part of a standard diagnostic interview, it is important to screen for current prescription medicines that may be associated with depressive symptoms (e.g., oral contraceptives, analgesics, proton pump

inhibitors, antihypertensives).^[9] It is also important to recognize that patients with bipolar disorder often present to health-care providers for the first time during a major depressive episode. As such, primary care physicians should assess those presenting with depression for a history of manic or hypomanic episodes. The treatment of bipolar depression requires different strategies than that of MDD; see [Bipolar Disorder](#) for more information.

Once a diagnosis of depression is confirmed, baseline and ongoing monitoring parameters are recommended; see [Table 2](#) for more information.

Table 1: Diagnostic Criteria of Major Depressive Disorder^[1]

Major depressive disorder is diagnosed when **5 or more of the following symptoms** occur during the same **2-week period**, representing a change from the patient’s usual functioning. **At least 1 of the symptoms must be either persistent depressed mood or marked loss of interest or pleasure.** Symptoms must be severe enough to interfere with daily life, including social relationships, work, school or other important activities.

Symptoms must not be substance-induced, caused by a medical condition, related to schizophrenia or other psychoses, and there must be no history of a manic or hypomanic episode (unless substance-induced or due to another medical condition).

When symptoms arise in the context of significant life stressors, clinical judgment is required to determine whether criteria for major depressive disorder are met.

Symptom	Description
Depressed mood	Persistent low or depressed mood for most of the time, almost daily; in children and adolescents, mood disturbance may present primarily as irritability
Loss of interest or pleasure	Significant reduction of interest or pleasure in usual daily activities
Fatigue or low energy	Ongoing fatigue or low energy almost daily
Sleep disturbance	Difficulty initiating or maintaining sleep, early morning awakening or excessive sleep almost daily
Appetite or weight change	Clinically meaningful changes in appetite or body weight not due to intentional dieting; in children, failure to achieve expected weight gain

Psychomotor slowing or agitation	Observable slowing of movement or speech, or increased restlessness or agitation almost daily
Impaired concentration or indecisiveness	Impaired concentration, thinking, or indecisiveness almost daily
Feelings of low self-worth or guilt	Disproportionate feelings of worthlessness or guilt almost daily
Thoughts of death or self-harm	Repeated thoughts about death; suicidal ideation (with or without a plan) or suicidal behaviour

Table 2: Monitoring Parameters for Depression^[1]

Assessment Type	Monitoring Parameter
Baseline assessments	Highly recommended • Conduct diagnostic work up/differential diagnosis including suicidality and organic causes of depression • Assess personal and family history including previous antidepressant use and alcohol, tobacco and substance use and dependence • Review physical health, including body mass index and (whenever deemed appropriate) waist circumference; metabolic syndrome; sexual health/dysfunction; hypertension; alcohol, tobacco and substance use and dependence Consider • Pregnancy test • Hemoglobin (anemia) • Thyroid-stimulating hormone • Liver function test Where indicated • Electrocardiogram for pre-existing cardiovascular disease or QTc interval prolongation • Bone density scan, especially when risk factors for osteoporosis are present • Electrolytes, especially in older patients
Ongoing assessments	Change compared to baseline

Assessment Type	Monitoring Parameter
	- Symptom changes, utilizing one of the assessment scales mentioned in Investigations - Weight and (where appropriate) waist circumference - Sexual dysfunction Treatment-emergent adverse effects - Suicidal thinking, especially in transitional youth - Elevated serum transaminases - Hyponatremia, especially in older patients - Hypertension; orthostatic hypotension
Special populations	Special considerations are required for children, older patients, patients during reproductive events, and those with concurrent mental or physical disorders

Modified with permission from Dodd S, Mitchell PB, Bauer M et al. Monitoring for antidepressant-associated adverse events in the treatment of patients with major depressive disorder: an international consensus statement. *World J Biol Psychiatry* 2018;19(5):330-48.

Therapeutic Choices

The main evidence for guideline-based treatment recommendations comes from studies of middle-aged patients; however, MDD occurs across the lifespan and presents specific challenges that need to be addressed in children and youth, people later in life, and patients in the perinatal and menopausal stages of life. The CANMAT 2023 update on clinical guidelines for MDD summarize the treatment protocol for adults.^[2] The CANMAT 2024 guidelines for perinatal mood and anxiety disorders summarize treatment for perinatal and postpartum depression.^[10] Updated guidelines for pediatric and geriatric populations are in development; until their publication, refer to the CANMAT 2016 clinical guidelines for MDD, which summarize the treatment protocol for each of these stages.^[12]

The most impressive evidence for the success of pharmacologic and depression-specific psychological therapies, in terms of both clinical efficacy and restoration of occupational functioning, comes from controlled trials involving multifaceted health-system interventions.^{[13][14]} These interventions, tested mostly in primary care settings, involve primary care physicians working in conjunction with another health-care provider (e.g., pharmacist, nurse, psychologist, psychiatrist); the prescriber initiates the antidepressants while the other practitioner educates the patient about depression and monitors the progress during treatment.^{[15][16]} Multifaceted interventions (nonpharmacologic and pharmacologic) can

substantially reduce long-term expense and increase the likelihood of the patient returning to work.^{[17][18]}^{[19][20][21]} Unfortunately not all clinicians and patients have access to these resources.

Nonpharmacologic Choices

Psychotherapy

While pharmacotherapy is preferable in severe depression and also when comorbid personality disorders are present, psychotherapies are as effective as medication for depression of mild to moderate severity.^[2] **Cognitive behavioural therapy (CBT), behavioural activation (BA) and interpersonal therapy (IPT)** are considered first-line recommendations for treating depression.^[22]

Each type of intervention has subtle advantages in specific circumstances, and patient preferences are important to consider. Interpersonal therapy may be preferable when major interpersonal issues are present. Newer versions of CBT and IPT have evolved and show particular promise: *mindfulness-based CBT* (mCBT) is particularly effective in preventing relapse into depression^[23] and *acceptance and commitment therapy* (ACT) has some efficacy in acute depression.^[24] It is also important to consider the significance of therapist cultural competence in delivering care to racialized groups.^[25]

Technology-Based Psychotherapy

Cognitive behavioural therapy is also available on the internet and through mobile applications, in some cases for a modest fee (e.g., [MindBeacon](#), [BounceBack](#), [moodgym](#) and [Headspace](#)).^{[2][26]} Multiple meta-analyses have demonstrated the efficacy of web-based CBT;^[27] unguided self-help may be less effective than guided self-help, particularly in individuals with moderate-to-severe depression.^{[28][29]} The 2023 CANMAT guidelines recommend adjunctive use of guided technology-based psychotherapies for mild-moderate depression. Guided options are preferred; unguided options are recommended only when supported with guidance by clinicians or when no other interventions are available.^[2] Patients should be encouraged to complete the modules and simple homework exercises on these sites in tandem with attending routine clinical care visits.

Motivational Interviewing

Motivational interviewing is a type of psychotherapy intended to improve the patient's motivation to change problematic behaviour. It is strikingly effective for treatment of substance abuse when depression is also present.^[22] Its effects are primarily on the substance abuse, not on the depressive symptoms.

Psychoeducation

In addition to psychotherapy, measures to enhance adherence to antidepressant therapy are useful. Psychoeducation with the following 5 simple messages is effective:^[30]

- Take medication daily.

- Call your prescriber or pharmacist for questions about side effects or other issues.
- Remember that it may take 2–4 weeks to see a noticeable effect from antidepressants.
- Continue to take medication even if you are feeling better.
- Do not stop taking your antidepressant without checking with your prescriber.

Other Nonpharmacologic Therapy

There is evidence for the benefit of regular physical **exercise** as monotherapy in mild depression and as an adjunct to pharmacotherapy and/or psychotherapy in moderate depression.^{[2][32]} Although there are few well-controlled studies of diet modification in the treatment of depression, a proof-of-concept RCT demonstrated feasibility of and positive outcomes for a dietary intervention that mainly reduced refined and processed foods while increasing healthier options (e.g., fruits, vegetables, lean proteins, nuts/seeds, olive oil).^[33] Adjunctive healthy diet or Mediterranean diet is recommended as a third-line lifestyle intervention in the latest CANMAT guidelines.^[2]

Light therapy is considered a first-line option for MDD with a seasonal pattern or as a second-line or adjunct measure for nonseasonal MDD of mild or moderate severity, respectively.^{[2][34][35]} In a randomized controlled trial, a combination of fluoxetine and light therapy demonstrated higher remission rates than either modality alone in patients with nonseasonal MDD.^[36] Light therapy involves exposure to a light box with 10 000-lux intensity slanted toward the face for 30 minutes per day, preferably in the morning. Improvement in depressive symptoms usually occurs within 1–3 weeks. Side effects of light therapy (e.g., headache, eye strain, irritability, insomnia) are usually mild and do not lead to treatment discontinuation. These can be minimized by having patients start with 10–15 minutes of exposure per day and gradually increase to 30 minutes per day. Patients with eye conditions that make them more vulnerable to light should check with their eye-care practitioner before starting light therapy. Light boxes with UV filters are recommended.

Novel neurostimulation therapies are proliferating as adjunctive strategies to treat depression. These include repetitive transcranial magnetic stimulation, magnetic seizure therapy and transcranial direct current stimulation, which have increasingly been evaluated but may not be routinely available.^{[37][38]}

Pharmacologic Choices

[Figure 1](#) illustrates an algorithm for the pharmacologic management of depression with or without a remission of symptoms.

Although diagnostic criteria for MDD and persistent depressive disorder (see [Table 1](#)) differ, recommendations for the use of **antidepressants** are the same in both conditions. [Table 3](#) lists first-, second- and third-line antidepressants according to the 2023 CANMAT guidelines.^[2] A network meta-analysis of 21 antidepressants involving over 100 000 patients confirmed superiority over placebo for all included agents and identified escitalopram and vortioxetine as the most effective and well-tolerated options available in Canada.^[40] [Table 6](#) provides more details about the drugs used to treat depression. A minimum therapeutic dose should be achieved in the first 2 weeks of treatment and increased if necessary

over the next 4–6 weeks. Patients need to be informed of common side effects and that some of these unwanted effects usually subside within 2 weeks (see [Table 4](#)). Consider switching within the same class during weeks 3–8 if a favourable response is overshadowed by persistent and troublesome side effects that do not respond to measures such as adjusting the dose or time of administration.

A systematic review of observational studies involving more than 200 000 adults in mid- and late life with moderate to severe depression found that exposure to SSRIs reduced the risk of suicide by more than 40% and 50% in the respective age groups.^[41] In contrast, regulatory authorities have issued cautions that highlight the potential of antidepressants to increase suicidal ideation, particularly in children and young adults. Clinicians are faced with complex decisions about when to use CBT and/or pharmacotherapy to treat depression in younger age groups and are advised to monitor patients carefully for any emergent suicidal ideation or acts, particularly during the early phase of treatment.^[42] The risk of emergent or worsening suicidal ideation must be weighed against the risks associated with untreated depression (suicide, increased duration and severity of illness, higher incidence of relapse).

Table 3: CANMAT Classification of Antidepressants^[2]

First-Line Agents ^[a]	bupropion	Second-Line Agents ^[a]	moclobemide
	citalopram		quetiapine
	desvenlafaxine		trazodone
	duloxetine		tricyclic antidepressants (amitriptyline, clomipramine, desipramine, doxepin, imipramine, nortriptyline, trimipramine)
	escitalopram		
	fluoxetine		
	fluvoxamine		
	levomilnacipran		
	mirtazapine		
	paroxetine		
	sertraline		
	venlafaxine	Third-Line Agents ^[a]	phenelzine
	vilazodone		tranylcypromine
	vortioxetine		

^[a] Within each category, antidepressants are listed in alphabetical order rather than in order of preference.

Selective Serotonin Reuptake Inhibitors (SSRIs)

SSRIs are first-choice antidepressants due to greater tolerability, ease of dosing and relatively low cost. Both time to onset (2–4 weeks) and rate of response (60–70%) are comparable to tricyclic antidepressants (TCAs); side effects mainly affect the GI tract, the CNS and sexual function.^[39] SSRIs can increase the risk of GI bleeding, particularly in patients with additional risk factors such as concomitant NSAID therapy or a history of GI bleeding. They should be used with caution in patients already at higher risk of GI bleeding.^[43] Unlike GI or CNS side effects, sexual dysfunction is more likely to persist during SSRI therapy and can involve impairment of desire, arousal and/or orgasm/ejaculation; there are also unsubstantiated reports of persisting sexual dysfunction following discontinuation of SSRIs/SNRIs.^[44] Consider the importance of sexual function to the patient when prescribing an antidepressant. Non-SSRI antidepressants that cause less sexual dysfunction (e.g., bupropion, mirtazapine, moclobemide) may be more acceptable for some patients.^[39]

Six SSRIs are currently available in Canada (**citalopram**, **escitalopram**, **fluoxetine**, **fluvoxamine**, **paroxetine** and **sertraline**). Escitalopram, the stereoisomer of citalopram, has a similar side effect profile but superior efficacy to citalopram^[45] and is at least as effective as venlafaxine.^[46] With the exception of fluoxetine, all SSRIs are commonly associated with discontinuation effects when stopping therapy. See [Antidepressant Discontinuation Syndrome](#) for more information. These effects can be severe and protracted in a small proportion of patients. Patients should be assessed regularly during and after treatment discontinuation for withdrawal symptoms and relapse.

Serotonin-Norepinephrine Reuptake Inhibitors (SNRIs)

Venlafaxine has inhibitory effects on serotonin reuptake at any therapeutic dose. At doses of >150 mg daily, it also inhibits norepinephrine reuptake. Compared with SSRIs, rates of remission have been reported to be 6–10% higher with venlafaxine, although these results are mainly based on comparisons with fluoxetine.^[47]^[48] Dose-related hypertension occurs rarely and is more likely to be seen at doses ≥225 mg. Monitor BP and avoid use in patients with uncontrolled hypertension.

Desvenlafaxine is the active metabolite of venlafaxine and may have a decreased risk of serious drug interactions. In the treatment of depression, the response and remission rates are comparable to venlafaxine. Desvenlafaxine is one of the few antidepressants to have been systematically evaluated and found to be efficacious in peri- and postmenopausal patients.^[49] Common adverse effects include insomnia, somnolence, dizziness and nausea.^[50]^[51]

Duloxetine appears to exert effects on both serotonin and norepinephrine systems at the starting dose of 60 mg daily and has comparable efficacy to venlafaxine.^[52] In addition to depression, duloxetine is indicated for neuropathic pain and for pain associated with fibromyalgia. Isolated cases of hepatic injury have been reported but there is no requirement for routine monitoring of liver enzymes. It is primarily metabolized by CYP1A2, of which tobacco smoke is a potent inducer; therefore, people who smoke cigarettes are likely to have lower plasma levels and may require higher dosing of duloxetine.^[53]

Levomilnacipran has higher potency for norepinephrine than serotonin reuptake inhibition (2:1). In the treatment of depression, the response and remission rates were 45% and 28%, respectively. Common

adverse effects include nausea, headache, dry mouth, hyperhidrosis, constipation and dizziness. Blood pressure and heart rate may increase and should be carefully monitored.^[54]

Dual Action Antidepressants

Bupropion is thought to exert antidepressant effects through norepinephrine and dopamine reuptake inhibition.^[55] It is a first-line agent for MDD and is also indicated for smoking cessation under a different brand name.^[56] Bupropion lowers the seizure threshold in a dose-dependent manner. It is contraindicated in patients with a seizure disorder or recent history of anorexia or bulimia nervosa, and should be used with extreme caution where there is a history of head trauma or seizure. Patients should be aware that the use of the medication with alcohol may lower seizure threshold. Treatment-emergent GI disturbance and sexual dysfunction occurs less frequently with bupropion compared with SSRIs.^[57]

Mirtazapine acts on both the noradrenergic and serotonergic systems. It has a lower rate of GI and sexual side effects but is associated with sedation and weight gain.^[58]

Other Serotonin Modulators

Trazodone is a potent postsynaptic serotonin (5-HT₂) receptor antagonist with weak serotonin reuptake inhibitory effects.^[59] Because of severe daytime sedation, trazodone is rarely prescribed at therapeutic antidepressant doses (300–400 mg daily) but is often prescribed at lower doses (50–100 mg) as a hypnotic in combination with other antidepressants.^[60] Tolerance, as seen with benzodiazepines, does not develop with trazodone and there is the potential for trazodone to enhance antidepressant effects.

Vortioxetine is a novel serotonergic antidepressant;^[61] its mechanism of action is thought to involve a combination of direct 5-HT receptor modulation and serotonin transporter (SERT) inhibition. In patients with MDD, vortioxetine has also shown positive effects on neuropsychological performance in multiple cognitive domains.^[62] In clinical trials, the most common side effects were GI-related. Mild to moderate nausea, typically in the first week of treatment, is usually self-resolving. The incidence of treatment-emergent sexual dysfunction and disrupted sleep may be lower with vortioxetine compared with SSRIs.^[63] However, patients should still be monitored for these adverse effects.

Vilazodone is a multimodal antidepressant that acts as a serotonin reuptake inhibitor and a partial agonist at 5-HT_{1A}. Vilazodone should be taken with food to ensure adequate absorption and a titration schedule is recommended to avoid GI effects. Nausea, diarrhea and headache are the most prominent adverse effects. Sexual side effects occur less frequently with vilazodone compared to SSRIs.

Tricyclic Antidepressants

Tricyclic antidepressants (**TCAs**) are generally reserved as second-line medications; their mechanism involves inhibition of the reuptake of norepinephrine and/or serotonin to varying degrees. **Amitriptyline** is the most frequently prescribed TCA, often at considerably lower doses of 25–50 mg per day for nighttime sedation or analgesia. **Nortriptyline**, a metabolite of amitriptyline and a secondary amine, has been used to treat depression in older populations for many years. **Clomipramine**, the most serotonergic TCA, is still

favoured in the treatment of obsessive compulsive disorder. In all cases, TCA use is limited by tolerability and safety concerns, especially cardiotoxicity following overdose.

Monoamine Oxidase Inhibitors

Prescribing irreversible MAOIs phenelzine and tranylcypromine is generally reserved for specialized mood disorder clinics because these agents are associated with potentially fatal food and drug interactions (hypertensive crisis, serotonin syndrome). Nevertheless, all prescribers and pharmacists should be aware of these risks and how they can be avoided when faced with a patient taking an irreversible MAOI. Despite the potential risks, under careful conditions, MAOIs may prove effective where other antidepressants have failed.

Moclobemide, a reversible and selective MAO-A inhibitor, does not require dietary restrictions when prescribed within the recommended dose range. Moclobemide is a well-tolerated alternative to SSRI or SNRI agents, particularly in patients with a significant anxiety component to their depressive episode. Although moclobemide is often perceived as being less effective than irreversible MAOIs, this has not been substantiated in clinical trials.

Antidepressant Adverse Effects

Table 4: Management of Treatment Emergent Adverse Effects of Antidepressant Therapy

The adverse effects listed in this table may occur with all antidepressants unless otherwise stated. The individual agents identified below are among the more commonly used antidepressants that have the highest prevalence of the listed adverse effect. Incidence rates are included when available.

General management: If adverse effects are severe, persist for longer than 2 weeks or are intolerable to the patient, consider (1) lowering dose or (2) switching agent.

Adverse Effect	Management	Prevalence
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GI upset (nausea, constipation, diarrhea)	Usually diminishes after 1–2 wk of therapy Nausea: once-daily dosing; take with food; have small, frequent meals; avoid strong smells Constipation: increase fibre intake, fluids, laxatives, exercise Diarrhea: have small, frequent meals; take probiotics; limit caffeine, alcohol, spicy food; fluid replacement	Nausea: fluvoxamine (37%), venlafaxine (37%), paroxetine (26%), sertraline (26%), vilazodone (24%), vortioxetine (23%), citalopram (21%), desvenlafaxine (22%), fluoxetine (21%), duloxetine (20%) Constipation: fluvoxamine (18%), venlafaxine (15%), paroxetine (14%), duloxetine (11%) Diarrhea: vilazodone (29%), sertraline (18%), paroxetine (11%)
Activation (anxiety, nervousness, agitation)	Usually diminishes after 1–2 wk of therapy Consider short-term use of benzodiazepines only for extreme anxiety; avoid use whenever possible. Severe agitation could be indicative of intolerance to SSRI therapy; switch to a different drug class	Nervousness: fluoxetine (14%), venlafaxine (10%) Anxiety: fluoxetine (12%) Agitation: fluvoxamine (16%)
Somnolence	Usually diminishes after 1–2 wk of therapy Bedtime dosing	Mirtazapine (54%), fluvoxamine (26%), paroxetine (23%), venlafaxine (17%), fluoxetine (13%), sertraline (13%)
Insomnia	Usually diminishes after 1–2 wk of therapy Daytime dosing Consider short-term use of hypnotic (e.g., benzodiazepines and Z-drugs) only for severe cases; avoid use whenever possible. Lemborexant may be considered	Venlafaxine (17%), bupropion (16%), fluoxetine (16%), sertraline (16%), fluvoxamine (14%), paroxetine (13%), duloxetine (11%)

	Counsel on sleep hygiene	
Weight gain	Antidepressants are associated with >5% weight gain, sustained over >5 y Counsel on healthy diet and exercise routine, monitor weight and metabolic parameters (BP, lipid, glucose) Consider switching to bupropion, which is usually associated with weight loss	Amitriptyline (>30%), citalopram (>2%), duloxetine (>2%), fluoxetine (>2%), mirtazapine (>30%), paroxetine (>10%), sertraline (>2%), trazodone (>2%), venlafaxine (>2%)
Sexual dysfunction	Improvement in sexual function usually occurs in patients who achieve remission of depression Consider starting PDE5 inhibitor in male patients Consider switching to antidepressants least associated with sexual dysfunction: bupropion, mirtazapine, vilazodone, vortioxetine	All SSRIs and SNRIs are associated with a high risk of sexual dysfunction Limited evidence suggests paroxetine may have the highest prevalence Risk of erectile dysfunction may be higher with SNRIs vs. SSRIs^[71]
Serotonin syndrome (dilated pupils, hyperreflexia, sweating, fever, agitation, nystagmus, clonus, delirium)	Stop the drug(s) and refer patient to the hospital	Rare, and may occur when multiple serotonergic agents are used MAOIs have the highest risk and require a minimum 2-wk washout period before another serotonergic agent is initiated Amitriptyline, mirtazapine and trazodone are unlikely to cause serotonin syndrome

Abbreviations:

MAOI = monoamine oxidase inhibitor; PDE5 = phosphodiesterase 5; SNRI = serotonin norepinephrine reuptake inhibitor; SSRI = selective serotonin reuptake inhibitor

Antidepressant Discontinuation Syndrome

Rapid discontinuation or dose reduction of most antidepressants may be associated with discontinuation syndrome.^[72] This syndrome is often reported with SSRIs, likely because these drugs are the most frequently prescribed, but any antidepressant taken for 6 weeks or more may be associated with discontinuation-related symptoms. Patients taking agents with a shorter half-life (e.g., paroxetine, venlafaxine) are at greatest risk of discontinuation-emergent symptoms, which include anxiety, crying, headache, increased dreaming, insomnia, irritability, myoclonus, nausea, electric shocks, tremor, flu-like symptoms, imbalance and sensory disturbance.

Patients experience somatic, neurologic and psychological symptoms attributed to a rapid decrease in the availability of 5-HT^[73] within 1–7 days of stopping the drug. If untreated, symptoms typically last between 3 days and 3 weeks, but may occasionally persist for several months.^[74]

Inform patients that discontinuation syndrome may occur if they abruptly stop or reduce the dose of their medication. Taper antidepressant doses gradually by approximately 25% per week and monitor for a re-emergence of depressive symptoms.^[72] Fluoxetine has a long half-life and can be tapered more rapidly than other SSRIs.

Reassure patients who experience discontinuation syndrome that the condition is not serious or life-threatening and severe symptoms will usually resolve in 3 days or less. The syndrome can be reversed by restarting the antidepressant and tapering the dose more slowly. Alternatively, if a slow taper is poorly tolerated, substitute with 1 dose of fluoxetine 10–20 mg PO. If discontinuation-emergent symptoms have not resolved after several days, additional doses of fluoxetine 20 mg may be taken if necessary; a total of 3 doses spread over 7–10 days is considered acceptable. Consider a herbal product containing ginger if drug interactions or adverse effects limit use of other possible anti-nausea medications (see [Nausea and Vomiting](#)).^[75]

There is some evidence that cognitive or mindfulness-based therapies combined with tapering may lessen the discontinuation-emergent effects.^{[76][77]}

Case reports have described patients with severe and persistent antidepressant discontinuation syndrome.^[2] It is hypothesized that these patients may have increased serotonin receptor sensitivity, resulting in exaggerated withdrawal symptoms due to small dose reductions. This requires dose reductions in increments smaller than commercially available (hyperbolic tapering), which necessitates compounding pharmacies' specialized preparation.^[78]

Atypical Antipsychotics (Serotonin and Dopamine Activity Modulators)

Atypical antipsychotics are serotonin and dopamine activity modulators, which can be an additional tool in the management of depression. Extended-release **quetiapine** is approved in Canada for the treatment of depression and is considered a second-line option.^{[39][79]} Immediate-release quetiapine is also used (off-label) for this indication but is associated with greater sedation.^[80]

Aripiprazole and **brexpiprazole** are approved as adjuncts to antidepressants in adults with MDD who have not had an adequate response to an antidepressant alone during the current episode. **Olanzapine** and **risperidone** are also used off-label as adjunctive therapy for treatment-resistant depression.

Response to antipsychotics should generally be observed within 2 weeks of initiation. See [Treatment-Resistant Depression](#) for information regarding the role of antipsychotics in the treatment of depression.

Natural Health Products

Several natural health products have been evaluated for the treatment of depression. It is important to consider potential drug interactions with these products since many should not be used in combination with prescription antidepressants. See [Table 6](#) for more information.

There is some evidence to support **St. John's wort** as a potential monotherapy option for patients with MDD of mild to moderate severity.^{[31][81]} As an inducer of CYP3A4 and P-glycoprotein, St. John's wort can decrease the bioavailability of many drugs. When combined with other serotonergic medications, there is an increased risk of serotonin syndrome. Consider potential interactions before recommending this agent.

S-adenosylmethionine (SAM-e), a synthetic form of a dietary amino acid, has been evaluated as adjunctive and monotherapy with limited evidence to support its use in MDD.^[81] SAM-e is well tolerated with few adverse effects, but may be associated with an increased risk of serotonin syndrome when combined with other serotonergic drugs.

Omega-3 fatty acids have been recommended as second-line therapy for the treatment of mild to moderate depression.^[31] However, a meta-analysis suggests that the evidence to support their use is of low quality and the benefits are not clinically significant.^[82]

Studies have looked at the efficacy of supplementation with **folic acid** and **vitamin D** for depression in adults. Folic acid was not effective when used to augment antidepressant medications;^[83] however, adjunctive **L-methylfolate**, a form of folate that crosses the blood–brain–barrier, 15 mg/day showed significantly greater efficacy as adjunctive therapy with SSRIs compared with SSRIs alone.^[84] There is preliminary evidence that L-carnosine, a polypeptide with antioxidant properties, may produce favourable antidepressant augmentation effects.^[85] Despite the low quality of evidence and need for more robust studies, a systematic review determined that vitamin D supplementation in adults with depression had no significant effect.^[86]

Rapid-Acting Therapies

Rapid-acting therapies are an emerging therapeutic class for the treatment of depression; these are treatments that yield improvement in symptoms within days and may be given for a short duration. They are part of a paradigm shift in the treatment of depression, particularly treatment-resistant depression. Examples include NMDA-modulators^[87] (**ketamine**, **esketamine**, **dextromethorphan/bupropion**^[88]), GABA_A-modulators (**zuranolone**^[89]) and the Stanford TMS protocol. Ketamine and esketamine are available in Canada; zuranolone has been recently approved but is not yet marketed; dextromethorphan/bupropion is not yet approved in Canada.

There is considerable interest in using NMDA receptor antagonists IV **ketamine** and intranasal **esketamine** as rapidly acting interventions for treatment-resistant depression.^[100] This represents a new avenue of therapeutic exploration, highlighting the relevance of the glutamatergic system to treatment-resistant depression (TRD).

Intranasal esketamine is approved in Canada for the treatment of TRD in combination with an SSRI or SNRI.^[101] It is available only through a controlled distribution program to ensure that all involved parties are trained on proper administration and monitoring.

Duration of Treatment—Relapse Prevention

Experts recommend the use of maintenance antidepressant therapies for a minimum period of 6–12 months, particularly when the intervals between depressive episodes become shorter and the disability associated with each depressive episode worsens. After achieving remission in a first episode, treat for 6–12 months. Patients may be at risk of recurrence if they have psychiatric or other comorbidities; residual symptoms; or if episodes are frequent, recurrent, chronic, severe or difficult to treat (see [Figure 1](#)). In these cases, treat for at least 2 years. Long-term maintenance treatment may also be considered.^[90]

In general, taper antidepressants slowly to minimize the risk of discontinuation-emergent symptoms. See [Antidepressant Discontinuation Syndrome](#) for more information.

Cognitive behavioural therapy and IPT may also be effective in reducing the risk of relapse.^{[22][91]}

Treatment-Resistant Depression

The current concept of TRD emphasizes failure to respond after 2 or more treatments of adequate dose and duration, assuming that medications were prescribed at optimal doses and for adequate duration prior to diagnosis.^[92] However, the definition may vary according to the planned intervention, e.g., the definition of TRD for an individual to receive CBT for TRD may be set at a lower threshold than for deep brain stimulation. Furthermore, there is a proposal to change the term to *difficult-to-treat depression*, as TRD may imply a resistance by the patient.^[93]

Antidepressants can be *switched* either within a medication class or to a different class. Most physicians switch out of class, e.g., from SSRI to SNRI or TCA, when there has been no response to the first drug.

Consider switching within a class in the presence of a favourable clinical response hampered by persistent, unmanageable side effects. For more information on switching between antidepressants see [Table 5](#).

Table 5: Switching Antidepressants

Switching From	Switching To	Generally Recommended Washout Period ^[a]
Most antidepressants	Most antidepressants	Generally there is no need for a washout period; a crossover technique can be applied whereby the dose of the first agent is tapered while the dose of the new antidepressant is gradually increased. Clomipramine is generally not recommended in cross-tapers; other exceptions are listed below. For specific cross-tapering instructions, consult specialized resources such as SwitchRx.ca
Any other antidepressant	Irreversible MAOI or moclobemide	5 half-lives of first antidepressant (3 wk for clomipramine and imipramine)
Irreversible MAOI	Any other antidepressant	2 wk (3 wk for clomipramine or imipramine)
Moclobemide	Any other antidepressant	5 days
Fluoxetine	Irreversible MAOI or moclobemide	5–6 wk
Fluoxetine	Any other antidepressant	Use caution after fluoxetine discontinuation due to its long half-life; a washout period is often recommended, depending on the new agent

Abbreviations:

MAOI = monoamine oxidase inhibitor

^[a] Recommendations are for general guidance only. For instructions specific to a patient scenario, consult specialized references such as [SwitchRx.ca](#).

Adjunctive therapy (adding a medication) is recommended when a patient tolerates the first antidepressant but has achieved only a partial response.^[39] *Combination* therapy (starting 2 medications at once) should be strongly considered for moderate-severe depression or refractory depression, particularly in individuals with a history of more than 3 previous depressive episodes.^{[94][95][96][98][99]} First-line options include aripiprazole and brexpiprazole.^[2] Second-line options include bupropion, ketamine, esketamine, olanzapine, quetiapine, risperidone, cariprazine, mirtazapine, modafinil, lithium and triiodothyronine.^[2] Third-line options include nortriptyline and lisdexamfetamine.^[2] Antipsychotic adjunctive therapy is not

recommended for long-term use, and close monitoring for movement disorders, weight gain and cardiometabolic effects is indicated. There is some evidence for increased mortality risk, even in younger patients who receive adjunctive therapy with atypical antipsychotics.^[97]

Among nonpharmacologic options, there is evidence to support adjunctive **CBT** in treatment-resistant depression.^[102] Neurostimulation therapies such as **electroconvulsive therapy** (ECT) and **repetitive transcranial magnetic stimulation** (rTMS) may be used in the management of treatment-resistant depression;^[2] however, a discussion about their use is beyond the scope of this chapter.

Choices during Pregnancy and Breastfeeding

Depression during Pregnancy

During pregnancy, symptoms of depression are common and can be challenging to disentangle from symptoms frequently reported during pregnancy (fatigue, sleeplessness, appetite changes). Perinatal depression is one of the most common complications of pregnancy^[103] and is a risk factor for suicide and infanticide.^{[104][105]} Pregnant patients with a history of depression or with current depressive symptoms should receive preventative interventions such as counselling.^[106]

The decision to initiate or continue antidepressant treatment during pregnancy should be based on a risk-benefit analysis that considers both the fetus and the pregnant patient. Untreated depression in pregnant patients is associated with untoward effects for the fetus/neonate, patient and others.^{[107][108][109][110][111]} If untreated, symptoms of depression may continue into the postpartum period or reoccur as postpartum depression.^{[112][113]}

Patients with a history of severe or recurrent depression who discontinue antidepressants during pregnancy may be at increased risk of relapse.^[114] Discontinuing antidepressants during pregnancy may increase risk of exacerbated anxiety and psychiatric emergency during the perinatal period.^{[115][116]}

Management

If symptoms are mild or if criteria for a mild-moderate depressive episode are met, psychotherapy (such as IPT and CBT) is the first treatment option.^{[10][117]} For moderate-severe episodes, antidepressant medication should be considered first line, especially in patients with a past history of depression.^{[10][118]} Although a dose-dependent relationship has not been elucidated, if antidepressants are prescribed during pregnancy, start at the lowest effective dose. As this area is actively evolving, clinicians are advised to consult a drug information service for the most up-to-date information (e.g., [MotherToBaby.org](#)).

Risks of Antidepressants During Pregnancy

Evidence suggests that SSRIs are not major teratogens. Although a statistical association may exist between cardiovascular malformations (e.g., septal defects) and fetal exposure to SSRIs, recent studies controlling for confounding factors do not support this association.^{[119][120][121]} SSRI exposure late in pregnancy may be associated with an increased risk of persistent pulmonary hypertension in the newborn.^{[122][123][124]} Paroxetine and fluoxetine may have higher associated risks than other SSRIs; see [Pharmacologic Choices During Pregnancy](#).

Poor neonatal adaptation syndrome is typically transient neonatal withdrawal and/or toxicity after exposure to antidepressants in utero. Signs and symptoms include tremulousness, irritability, increased muscle tone, sleeping disturbances and respiratory distress; more severe symptoms have also been reported.^{[125][126]} Although reports of incidence vary from 5–85%, likely due to lack of a standardized definition, up to about 30% is typical.^[127]

Adverse delivery outcomes (e.g., preterm birth, low birth weight) have also been statistically associated with antenatal antidepressant use, although the clinical relevance is not certain.^{[128][129][130]}

The relationships between perinatal exposure to antidepressants and incidence of autism and attention deficit hyperactivity disorder have been a topic of great debate; however, the evidence is not strong enough to draw clear conclusions. Studies examining the effect of antidepressant exposure on growth and development of brain, language and cognition also have inconsistent findings that are confounded by maternal psychopathology.^[131] The Canadian Paediatric Society 2021 position statement summarizes the evidence and concludes that overall “...the risk of serious adverse effects from exposure to SSRIs or SNRIs in utero is low.”^[132]

With regards to risks to the pregnant patient, antidepressants have been associated with maternal hypertension/pre-eclampsia, postpartum hemorrhage and caesarean section.^{[133][134]} Studies, however, have not been consistent in their findings.^[135]

Pharmacologic Choices During Pregnancy

If pharmacologic therapy is indicated, **citalopram**, **escitalopram** and **sertraline** are first-line options.^[10] Fluoxetine is a second-line option since its association with major malformations is more controversial.^[10]^[119] Paroxetine has been associated with risk for major malformations, and although the clinical magnitude is questionable, it is still not recommended for perinatal depression, unless the patient was stable on it prior to conception.^{[10][136][137]}

Second-line options in the treatment of perinatal depression include bupropion, desvenlafaxine, duloxetine, fluoxetine, fluvoxamine, mirtazapine, and venlafaxine.^[10] MAOIs and doxepin should be avoided during pregnancy. Clomipramine should be considered only if the patient was stabilized on it prior to conception. Other antidepressants have not been studied sufficiently in pregnancy to make recommendations on their use (e.g., vilazodone, vortioxetine, levomilnacipran).^[10]

Postpartum Mood Disturbances

Postpartum mood disturbances is a term that encompasses the following 3 presentations: postpartum “blues,” which affect up to 80% of patients after a pregnancy,^[138] postpartum major depression, which occurs in up to 16% of patients in Canada after a pregnancy and postpartum psychosis, which occurs in $\leq 0.1\%$.^{[139][140][141]}

The blues are not a disorder per se, given that they are so common and symptoms are self-limiting, requiring only monitoring and supportive care. However, patients with postpartum blues are at risk of developing a depressive episode.

Management

Psychotherapy, preferably IPT or CBT, should be considered first for the treatment of postpartum depression, particularly if the patient is breastfeeding. When the illness is more severe, psychotherapy may be inadequate or inappropriate, in which case antidepressant treatment should be initiated. For breastfeeding patients, the 2024 CANMAT guidelines on perinatal depression recommend consideration of **sertraline**, **escitalopram** or **citalopram** as first-line pharmacologic options.^[10] Second-line options include bupropion, duloxetine, desvenlafaxine, venlafaxine, fluoxetine, fluvoxamine and mirtazapine. Third-line options include paroxetine, trazodone and TCAs other than doxepin. Other antidepressants lack sufficient data in breastfeeding patients. Doxepin is not recommended due to documented significant adverse effects in the breastfed infant. **Zuranolone** is a GABA_A-modulator recently approved by Health Canada but not yet available.^[89]

A discussion of general principles on the use of medications in these special populations can be found in [Drugs Use during Pregnancy](#) and [Drug Use during Breastfeeding](#). Other specialized reference sources are also provided in these appendices.

Therapeutic Tips

- Prescribers should choose 1 or 2 agents from several antidepressant classes and use them consistently to acquire expertise in the use of those medications.
- Individualize therapy by considering patient comorbidities and most significant symptoms of depression (e.g., sleep disturbances, cognitive dysfunction, anxiety, somatic symptoms); see the [Antidepressant Decision Support Tool](#) developed in 2022 to help guide selection.
- Before prescribing an antidepressant, discuss side effect profiles, drug-drug interactions and discontinuation-emergent side effects with the patient.
- Provide key psychoeducational messages (as noted in [Nonpharmacologic Choices](#)) with the initial prescription. Reinforce and add to this during regular follow-up visits.
- A combination of pharmacotherapy and psychotherapy (specifically IPT or CBT) is superior to either modality alone. In mild to moderate cases, combination therapy provides little synergy compared to either

treatment alone, but does provide protection against early relapse compared to either treatment alone. Pharmacotherapy typically facilitates more rapid relief of symptoms, but does not prevent relapse if the medication is stopped; both IPT and CBT reduce relapse for up to 3 years after completion of a course of psychotherapy.

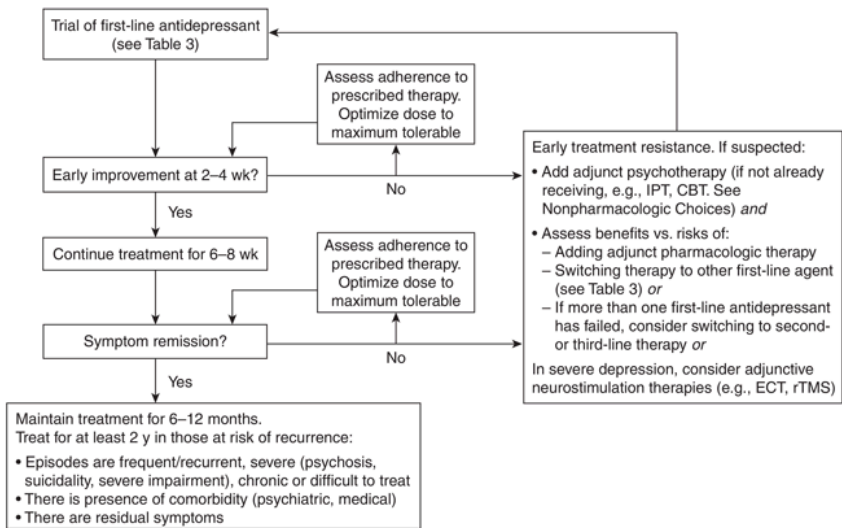
- Reinforce the importance of maintenance therapy beyond the acute phase.
- There is a limited role for therapeutic drug-level monitoring as there is a poor correlation between serum levels of SSRI or SNRI antidepressants and clinical response or side effects. Similarly, there is only a limited benefit in performing pharmacogenetic testing for cytochrome P450 enzyme polymorphisms. Pharmacogenetic testing is, however, warranted in cases such as a failure to respond to high doses of several antidepressants (which may suggest that the patient is an ultra-rapid metabolizer) or repeated inability to tolerate low doses of several agents (which may suggest a poor metabolizer status).
- In nonresponders, review alcohol and drug abuse history, assess medication adherence and confirm diagnosis.
- Refer for psychiatric consultation if the patient has psychotic symptoms or acute suicidal ideation, or after failure of 3 treatment trials.
- When discontinuing antidepressants, taper slowly over 4–6 weeks. This is particularly important for paroxetine and venlafaxine.

Resources

General patient education: [CANMAT](#), [moodgym](#), or self-help books such as *Mind Over Mood*^[142] or *Feeling Good*.^[143] Patients may also consult [The Choice-D Patient and Family Guide to Depression Treatment](#), which translates the CANMAT MDD guidelines into patient-friendly advice.

Algorithms

Figure 1: Pharmacologic Treatment of Depression^[39]



Abbreviations: CBT = cognitive behavioural therapy; ECT = electroconvulsive therapy; IPT = interpersonal psychotherapy; rTMS = repetitive transcranial magnetic stimulation

Drug Table

Table 6: Drug Therapy for Depression

Drug/Cost ^[a]	Dosage	Adverse Effects ^[b]	Drug Interactions
Drug Class: Antipsychotics, second-generation			
aripiprazole Abilify, generics < \$30	2–15 mg daily PO Indication: first-line adjunctive therapy	All antipsychotics: Advise patients about antipsychotic- associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection. Aripiprazole: EPS , dizziness, orthostatic	Strong inducers of CYP2D6 or CYP3A4 (e.g., carbamazepine, phenytoin, rifampin) can decrease aripiprazole levels substantially. Strong inhibitors of CYP2D6 or CYP3A4 (e.g., ketoconazole, quinidine, fluoxetine, paroxetine) can

Drug/Cost ^[a]	Dosage	Adverse Effects ^[b]	Drug Interactions
		hypotension, headache, GI complaints, nasopharyngitis, tremor, sedation, insomnia.	increase aripiprazole levels substantially.
brexpiprazole  Rexulti \$90–120	1–2 mg daily PO Indication: first-line adjunctive therapy	All antipsychotics: Advise patients about antipsychotic-associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection. Brexpiprazole: weight gain, hyperglycemia, elevated triglycerides, EPS (appears to be less common than with aripiprazole), headache, orthostatic hypotension (rare).	Strong inducers of CYP2D6 or CYP3A4 (e.g., carbamazepine, phenytoin, rifampin) can decrease aripiprazole levels substantially. Strong inhibitors of CYP2D6 or CYP3A4 (e.g., ketoconazole, quinidine, fluoxetine, paroxetine) can increase aripiprazole levels substantially.

Drug/Cost ^[a]	Dosage	Adverse Effects ^[b]	Drug Interactions
olanzapine Zyprexa , Zyprexa Zydis, generics < \$30	2.5–10 mg daily PO Indication: second-line adjunctive therapy	All antipsychotics: Advise patients about antipsychotic-associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection. Olanzapine: weight gain, dizziness, sedation, anticholinergic effects, hepatic aminotransferase elevation, orthostatic hypotension, increased risk of diabetes and dyslipidemia, EPS (especially akathisia).	Sedation with CNS depressants; may potentiate antihypertensive drug effects; inhibitors of CYP1A2 or CYP2D6 (e.g., diltiazem, fluvoxamine, paroxetine) may increase olanzapine levels; inducers of CYP1A2 or CYP2D6 (e.g., barbiturates, carbamazepine, phenytoin, rifampin or cigarette smoking) may decrease olanzapine levels.

Drug/Cost ^[a]	Dosage	Adverse Effects ^[b]	Drug Interactions
quetiapine, extended-release Seroquel XR, generics < \$30	Initial: 50 mg daily PO for 2 days, then 150 mg daily; if necessary, can increase to 300 mg on or after day 4 Usual: 150 mg/day PO High: 300 mg/day PO Indication: low doses (25–100 mg) can be used to augment standard antidepressants (second line). Higher doses (150–600 mg) may be prescribed as monotherapy or in combination based on tolerability	All antipsychotics: Advise patients about antipsychotic-associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection. Quetiapine: sedation, dizziness, weight gain, orthostatic hypotension, hepatic transaminase elevation, headache, anticholinergic effects, increased risk of diabetes and dyslipidemia, possible increased risk of cataracts; may reduce thyroid hormone levels.	Sedation with CNS depressants; may potentiate antihypertensive drug effects; inhibitors of CYP1A2 or CYP2D6 (e.g., diltiazem, fluvoxamine, paroxetine) may increase olanzapine levels; inducers of CYP1A2 or CYP2D6 (e.g., barbiturates, carbamazepine, phenytoin, rifampin or cigarette smoking) may decrease olanzapine levels.
risperidone  Risperdal , generics < \$30	0.25–2 mg daily PO Indication: second-line adjunctive therapy	All antipsychotics: Advise patients about antipsychotic-associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection. Risperidone: sedation, headache, weight gain, orthostatic hypotension, rhinitis, anxiety, dose-related hyperprolactinemia, EPS . Risk of intraoperative floppy iris syndrome in	Additive sedation with CNS depressants; may potentiate antihypertensive drug effects; inhibitors of CYP3A4 (e.g., clarithromycin, erythromycin, grapefruit juice, ketoconazole, prednisone) may increase risperidone levels; inducers of CYP3A4 (e.g., carbamazepine, phenytoin, rifampin) may decrease risperidone levels.

Drug/Cost ^[a]	Dosage	Adverse Effects ^[b]	Drug Interactions
Drug Class: Dual Action Antidepressants			
bupropion  Wellbutrin SR, Wellbutrin XL, Olan Bupropion SR, other generics < \$30	SR formulation (doses >150 mg/day PO should be given in divided doses): Initial: [c] 150 mg/day PO Usual: 150–300 mg/day PO High: [d] 375–450 mg/day PO XL formulation (given once daily): Initial: [c] 150 mg/day PO Usual: 150–300 mg/day PO High: [d] 450 mg/day PO	patients undergoing cataract surgery who have been exposed to risperidone. Agitation, insomnia, anorexia; contraindicated in anorexia or bulimia nervosa and seizure disorders.	Use with MAOIs , linezolid or methylene blue may lead to potentially fatal reaction initially presenting with tremor, agitation, hypomania, hyperthermia and/or hypertension. May increase levels of cyclophosphamide, ifosfamide and orphenadrine.
mirtazapine  Remeron, Remeron RD, generics < \$30	Initial: [c] 15–30 mg/day PO Usual: 30–45 mg/day PO High: [d] 60 mg/day PO	Weight gain, sedation.	Use with MAOIs , linezolid or methylene blue may lead to potentially fatal reaction initially presenting with tremor, agitation, hypomania, hyperthermia and/or hypertension. Sedative effects may be potentiated by alcohol or

Drug/Cost ^[a]	Dosage	Adverse Effects ^[b]	Drug Interactions
			benzodiazepines. QT _c prolongation and torsades de pointes have occurred in patients at risk of QT _c prolongation, in patients taking concomitant medications that prolong QT _c and in cases of drug overdose. Orally disintegrating tablets can be taken without water.

Drug Class: Irreversible Monoamine Oxidase Inhibitors

phenelzine Nardil	Initial: ^[c] 15–30 mg/day PO	Edema, postural hypotension, insomnia, sexual dysfunction.	Many significant reactions reported. Increased risk of potentially fatal serotonin syndrome and/or hypertensive crisis. Consult reliable drug information reference; tyramine-containing food may also cause hypertensive crisis (see Resources).
\$30–60	Usual: 30–75 mg/day PO High: ^[d] 90–120 mg/day PO		

tranylcypromine  Parnate	Initial: ^[c] 10–20 mg/day PO	Edema, postural hypotension, insomnia, sexual dysfunction.	Many significant reactions reported. Increased risk of potentially fatal serotonin syndrome and/or hypertensive crisis. Consult reliable drug information reference; tyramine-containing food may also cause hypertensive crisis (see Resources).
\$30–60	Usual: 20–60 mg/day PO High: ^[d] 60–80 mg/day PO		

Drug Class: NMDA Receptor Antagonists

esketamine  Spravato	Initial: 28–56 mg intranasally twice	Anxiety, dissociation, sedation, headache,	Excess sedation with CNS depressants; may require
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Drug/Cost ^[a]	Dosage	Adverse Effects ^[b]	Drug Interactions
\$275/28 mg unit	weekly. Wk 1–4: may increase by 28 mg increments up to 84 mg twice weekly. Wk 5–8: decrease to once weekly. From wk 9: may decrease to Q2 wk Usual: 56–84 mg intranasally Q1–2 wk	dizziness, vertigo, hypoesthesia, dysgeusia, nausea, vomiting, increased BP.	dosage adjustments and/or closer monitoring. Alcohol consumption is not recommended within 24 h of administration. Risk of hypertensive crisis with other drugs that increase BP (e.g., amphetamine salts, MAOIs , methylphenidate, triptans); avoid combination or monitor BP closely. Avoid using other nasal sprays within 1 hr of administration.

Drug Class: Reversible Monoamine Oxidase-A Inhibitors

moclobemide generics	Initial: ^[c] 200–300 mg/day PO	Nausea, insomnia, dizziness.	Many significant reactions reported. Increased risk of potentially fatal serotonin syndrome and/or hypertensive crisis. Consult reliable drug information reference.
\$30–60	Usual: 450–600 mg/day PO High: ^[d] 900 mg/day PO		

Drug Class: Selective Serotonin Reuptake Inhibitors

Drug/Cost ^[a]	Dosage	Adverse Effects ^[b]	Drug Interactions
citalopram Celexa, CTP 30, other generics < \$30	Initial: ^[c] 10– 20 mg/day PO Usual: ^[f] 20– 40 mg/day PO High: ^[g] 40 mg/day PO	All SSRIs : Upon initiation: GI upset, anorexia, dry mouth, diaphoresis, headache, dizziness, insomnia, somnolence, anxiety, agitation, tremor. Usually resolve after 2 wk of therapy. Others: sexual dysfunction, weight gain, SIADH with hyponatremia. Citalopram: dose- related QTc prolongation.	SSRIs are all substrates of various CYP enzymes and subject to drug interactions. Concurrent use with MAOIs , linezolid and methylene blue is contraindicated due to increased risk of serotonin syndrome. Use caution if combining with other serotonergic agents. Increased risk of GI bleeding with NSAIDs , antiplatelet agents. Avoid concurrent use with drugs associated with prolonged QTc interval/torsades de pointes. Citalopram: higher risk of QTc prolongation, especially at high doses and in combination with other QTc- prolonging agents.
escitalopram Cipralex, KYE- Escitalopram, other generics < \$30	Initial: ^[c] 10 mg/day PO Usual: ^[f] 10– 20 mg/day PO High: ^[g] 20 mg/day PO	All SSRIs : Upon initiation: GI upset, anorexia, dry mouth, diaphoresis, headache, dizziness, insomnia, somnolence, anxiety, agitation, tremor. Usually resolve after 2 wk of therapy. Others: sexual dysfunction, weight gain, SIADH with hyponatremia. Escitalopram: dose- related QTc prolongation.	SSRIs are all substrates of various CYP enzymes and subject to drug interactions. Concurrent use with MAOIs , linezolid and methylene blue is contraindicated due to increased risk of serotonin syndrome. Use caution if combining with other serotonergic agents. Increased risk of GI bleeding with NSAIDs , antiplatelet agents. Avoid concurrent use with drugs associated with prolonged QTc interval/torsades de pointes. Escitalopram: higher risk of QTc prolongation, especially at high doses and in

Drug/Cost ^[a]	Dosage	Adverse Effects ^[b]	Drug Interactions
fluoxetine Odan-Fluoxetine, other generics < \$30	Initial: ^[c] 10– 20 mg/day PO Usual: ^[f] 20– 40 mg/day PO High: ^[g] 60– 80 mg/day PO	All SSRIs : Upon initiation: GI upset, anorexia, dry mouth, diaphoresis, headache, dizziness, insomnia, somnolence, anxiety, agitation, tremor. Usually resolve after 2 wk of therapy. Others: sexual dysfunction, weight gain, SIADH with hyponatremia. Fluoxetine: anxiety upon initiation is common.	combination with other QTc- prolonging agents. SSRIs are all substrates of various CYP enzymes and subject to drug interactions. Concurrent use with MAOIs , linezolid and methylene blue is contraindicated due to increased risk of serotonin syndrome. Use caution if combining with other serotonergic agents. Increased risk of GI bleeding with NSAIDs , antiplatelet agents. Avoid concurrent use with drugs associated with prolonged QTc interval/torsades de pointes. Fluoxetine: potent inhibitor of CYP2D6 .
fluvoxamine Luvox, generics < \$30	Initial: ^[c] 50– 100 mg/day PO Usual: ^[f] 150– 200 mg/day PO High: ^[g] 400 mg/day PO	All SSRIs : Upon initiation: GI upset, anorexia, dry mouth, diaphoresis, headache, dizziness, insomnia, somnolence, anxiety, agitation, tremor. Usually resolve after 2 wk of therapy. Others: sexual dysfunction, weight gain, SIADH with hyponatremia. Fluvoxamine: somnolence and nausea are common.	SSRIs are all substrates of various CYP enzymes and subject to drug interactions. Concurrent use with MAOIs , linezolid and methylene blue is contraindicated due to increased risk of serotonin syndrome. Use caution if combining with other serotonergic agents. Increased risk of GI bleeding with NSAIDs , antiplatelet agents. Avoid concurrent use with drugs associated with prolonged QTc interval/torsades de pointes. Fluvoxamine: potent inhibitor of CYP1A2 .

Drug/Cost ^[a]	Dosage	Adverse Effects ^[b]	Drug Interactions
paroxetine, immediate-release Paxil , generics < \$30	Initial: ^[c] 10–20 mg/day PO Usual: ^[f] 20–40 mg/day PO High: ^[g] 60 mg/day PO	All SSRIs : Upon initiation: GI upset, anorexia, dry mouth, diaphoresis, headache, dizziness, insomnia, somnolence, anxiety, agitation, tremor. Usually resolve after 2 wk of therapy. Others: sexual dysfunction, weight gain, SIADH with hyponatremia. Paroxetine: anticholinergic effects (dry mouth, constipation), somnolence are common.	SSRIs are all substrates of various CYP enzymes and subject to drug interactions. Concurrent use with MAOIs , linezolid and methylene blue is contraindicated due to increased risk of serotonin syndrome. Use caution if combining with other serotonergic agents. Increased risk of GI bleeding with NSAIDs , antiplatelet agents. Avoid concurrent use with drugs associated with prolonged QTc interval/torsades de pointes. Paroxetine: potent CYP2D6 inhibitor.
paroxetine, controlled-release Paxil CR \$60–90	Initial: ^[c] 12.5–25 mg/day PO Usual: ^[f] 25–50 mg/day PO High: ^[g] 75 mg/day PO	All SSRIs : Upon initiation: GI upset, anorexia, dry mouth, diaphoresis, headache, dizziness, insomnia, somnolence, anxiety, agitation, tremor. Usually resolve after 2 wk of therapy. Others: sexual dysfunction, weight gain, SIADH with hyponatremia. Paroxetine: anticholinergic effects (dry mouth, constipation), somnolence are common.	SSRIs are all substrates of various CYP enzymes and subject to drug interactions. Concurrent use with MAOIs , linezolid and methylene blue is contraindicated due to increased risk of serotonin syndrome. Use caution if combining with other serotonergic agents. Increased risk of GI bleeding with NSAIDs , antiplatelet agents. Avoid concurrent use with drugs associated with prolonged QTc interval/torsades de pointes. Paroxetine: potent CYP2D6 inhibitor.

Drug/Cost ^[a]	Dosage	Adverse Effects ^[b]	Drug Interactions
sertraline Zoloft , generics < \$30	Initial: ^[c] 25–50 mg/day PO Usual: ^[f] 50–100 mg/day PO High: ^[g] 150–200 mg/day PO	All SSRIs : Upon initiation: GI upset, anorexia, dry mouth, diaphoresis, headache, dizziness, insomnia, somnolence, anxiety, agitation, tremor. Usually resolve after 2 wk of therapy. Others: sexual dysfunction, weight gain, SIADH with hyponatremia.	SSRIs are all substrates of various CYP enzymes and subject to drug interactions. Concurrent use with MAOIs , linezolid and methylene blue is contraindicated due to increased risk of serotonin syndrome. Use caution if combining with other serotonergic agents. Increased risk of GI bleeding with NSAIDs , antiplatelet agents. Avoid concurrent use with drugs associated with prolonged QTc interval/torsades de pointes. Sertraline: weak CYP2D6 inhibitor.
Drug Class: Serotonin-Norepinephrine Reuptake Inhibitors			
desvenlafaxine  Pristiq , generics \$60–90	Initial: 50 mg daily PO Usual: 50 mg daily PO High: 100 mg daily PO	Nausea, sleep disturbance, drowsiness, nervousness, dizziness, dry mouth.	Use with MAOIs , linezolid or methylene blue may lead to potentially fatal reaction initially presenting with tremor, agitation, hypomania, hyperthermia and/or hypertension. Potent inhibitors of CYP3A4 may increase serum drug concentrations.
duloxetine Cymbalta , generics \$30–60	Initial: 60 mg daily PO Usual: 60 mg daily PO If necessary for tolerability, may start with 30 mg/day and increase to 60 mg in 1–2 wk	Nausea, drowsiness, insomnia, dizziness, dry mouth.	Use with MAOIs , linezolid or methylene blue may lead to potentially fatal reaction initially presenting with tremor, agitation, hypomania, hyperthermia and/or hypertension. Do not use with potent inhibitors of CYP1A2 such as

Drug/Cost ^[a]	Dosage	Adverse Effects ^[b]	Drug Interactions
	High: 120 mg/day PO		ciprofloxacin, fluvoxamine and ketoconazole.
levomilnacipran  Fetzima \$120–150	Initial: 20 mg/day PO for 2 days then 40 mg/day Usual: 40–120 mg/day PO	Nausea, dry mouth, constipation, hyperhidrosis, headache, dizziness, increased heart rate, tachycardia, insomnia, erectile dysfunction.	Use with MAOIs , linezolid or methylene blue may lead to potentially fatal reaction initially presenting with tremor, agitation, hypomania, hyperthermia and/or hypertension. Potent inhibitors of CYP3A4 may increase serum drug concentrations.
venlafaxine  Effexor XR , generics < \$30	Initial: [a] 37.5–75 mg/day PO Usual: 112.5–225 mg/day PO High: [a] 300–375 mg/day PO	Nausea, sleep disturbance, drowsiness, nervousness, dizziness, dry mouth. Dose-related hypertension occurs rarely, particularly at doses ≥225 mg/day.	Use with MAOIs , linezolid or methylene blue may lead to potentially fatal reaction initially presenting with tremor, agitation, hypomania, hyperthermia and/or hypertension. Strong inhibitors of CYP2D6 or CYP3A4 (e.g., ketoconazole, quinidine, fluoxetine, paroxetine) may increase venlafaxine levels.

Drug Class: Serotonin Modulators

trazodone generics < \$30	Initial: [a] 150–200 mg/day PO Usual: 300–400 mg/day PO High: [a] 600 mg/day PO	Drowsiness, orthostatic hypotension, nausea, headache, dry mouth, priapism.	Toxicity may be increased by inhibitors of CYP3A4 (e.g., clarithromycin, erythromycin, grapefruit juice, ketoconazole). Effectiveness may be reduced by inducers of CYP3A4 such as carbamazepine, phenytoin and rifampin.
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Drug/Cost ^[a]	Dosage	Adverse Effects ^[b]	Drug Interactions
			May potentiate effects of other CNS depressants and augment hypotensive effects of antihypertensives. Avoid use with MAOIs .
vilazodone Viibryd, generics \$90–120	Initial: [a] 10 mg/day PO with food for 7 days Usual: 20 mg/day PO with food; may increase to 40 mg/day after 7 days, if needed	Diarrhea, nausea, vomiting, insomnia.	Potent inhibitors of CYP 3A4 may significantly increase drug levels; do not use doses above 20 mg/day when coadministered with a strong inhibitor. Use with MAOIs , linezolid or methylene blue may lead to potentially fatal reaction initially presenting with tremor, agitation, hypomania, hyperthermia and/or hypertension. Use caution if coadministering with drugs that affect serotonergic neurotransmitter systems (e.g., dextromethorphan, fentanyl, lithium, meperidine, methadone, pentazocine, SSRIs , St. John's wort, tapentadol, tramadol, triptan, tryptophan).

vortioxetine Trintellix , generics \$30–60	Initial: [a] 5–10 mg daily PO Usual: [a] 10–20 mg daily PO	Nausea, constipation, vomiting. Transient symptoms associated with abrupt discontinuation include headache, increased dreaming, mood swings,	Use with MAOIs , linezolid or methylene blue may lead to potentially fatal reaction initially presenting with tremor, agitation, hypomania, hyperthermia and/or hypertension.
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Drug/Cost ^[a]	Dosage	Adverse Effects ^[b]	Drug Interactions
		muscle tension, vertigo, rhinorrhea.	Use caution if coadministering with drugs that affect serotonergic neurotransmitter systems (e.g., dextromethorphan, fentanyl, lithium, meperidine, methadone, pentazocine, SSRIs , St. John's wort, tapentadol, tramadol, triptan, tryptophan). Potential additive bleeding risk with drugs such as warfarin, ASA and other antiplatelet agents.

Drug Class: Tricyclic Antidepressants

amitriptyline generics	Initial: ^[c] 25–50 mg/day PO Usual: 75–200 mg/day PO High: ^[d] 250–300 mg/day PO	Anticholinergic (dry mouth, blurred vision, constipation, urinary hesitancy, tachycardia, delirium), antihistaminergic (sedation, weight gain), orthostatic hypotension, lowered seizure threshold; sexual dysfunction.	Use with MAOIs , linezolid or methylene blue may lead to potentially fatal reaction initially presenting with tremor, agitation, hypomania, hyperthermia and/or hypertension. Inducers of CYP1A2 (e.g., barbiturates, carbamazepine and rifampin) may decrease effect; cimetidine and antipsychotics may increase effect and toxicity; possible interaction with antiarrhythmics (may lead to increased effect of either drug); may reduce antihypertensive effect of clonidine; may augment hypotensive effect of thiazides.
< \$30			

Drug/Cost ^[a]	Dosage	Adverse Effects ^[b]	Drug Interactions
clomipramine Anafranil, generics	Initial: ^[c] 25–50 mg/day PO Usual: 75–200 mg/day PO High: ^[d] 250–300 mg/day PO	Anticholinergic (dry mouth, blurred vision, constipation, urinary hesitancy, tachycardia, delirium), antihistaminergic (sedation, weight gain), orthostatic hypotension, lowered seizure threshold; sexual dysfunction.	Use with MAOIs , linezolid or methylene blue may lead to potentially fatal reaction initially presenting with tremor, agitation, hypomania, hyperthermia and/or hypertension. Inducers of CYP1A2 (e.g., barbiturates, carbamazepine and rifampin) may decrease effect; cimetidine and antipsychotics may increase effect and toxicity; possible interaction with antiarrhythmics (may lead to increased effect of either drug); may reduce antihypertensive effect of clonidine; may augment hypotensive effect of thiazides.
\$30–60			
desipramine generics	Initial: ^[c] 25–50 mg/day PO Usual: 75–200 mg/day PO High: ^[d] 250–300 mg/day PO	Anticholinergic (dry mouth, blurred vision, constipation, urinary hesitancy, tachycardia, delirium), antihistaminergic (sedation, weight gain), orthostatic hypotension, lowered seizure threshold; sexual dysfunction.	Use with MAOIs , linezolid or methylene blue may lead to potentially fatal reaction initially presenting with tremor, agitation, hypomania, hyperthermia and/or hypertension. Inducers of CYP1A2 (e.g., barbiturates, carbamazepine and rifampin) may decrease effect; cimetidine and antipsychotics may increase effect and toxicity; possible interaction with antiarrhythmics (may lead to increased effect of either drug); may reduce antihypertensive effect of
\$30–60			

Drug/Cost ^[a]	Dosage	Adverse Effects ^[b]	Drug Interactions
			clonidine; may augment hypotensive effect of thiazides.
doxepin Sinequan \$60–90	Initial: ^[a] 25–50 mg/day PO Usual: 75–200 mg/day PO High: ^[a] 250–300 mg/day PO	Anticholinergic (dry mouth, blurred vision, constipation, urinary hesitancy, tachycardia, delirium), antihistaminergic (sedation, weight gain), orthostatic hypotension, lowered seizure threshold; sexual dysfunction.	Use with MAOIs , linezolid or methylene blue may lead to potentially fatal reaction initially presenting with tremor, agitation, hypomania, hyperthermia and/or hypertension. Inducers of CYP1A2 (e.g., barbiturates, carbamazepine and rifampin) may decrease effect; cimetidine and antipsychotics may increase effect and toxicity; possible interaction with antiarrhythmics (may lead to increased effect of either drug); may reduce antihypertensive effect of clonidine; may augment hypotensive effect of thiazides.
imipramine generics \$30–60	Initial: ^[a] 25–50 mg/day PO Usual: 75–200 mg/day PO High: ^[a] 250–300 mg/day PO	Anticholinergic (dry mouth, blurred vision, constipation, urinary hesitancy, tachycardia, delirium), antihistaminergic (sedation, weight gain), orthostatic hypotension, lowered seizure threshold; sexual dysfunction.	Use with MAOIs , linezolid or methylene blue may lead to potentially fatal reaction initially presenting with tremor, agitation, hypomania, hyperthermia and/or hypertension. Inducers of CYP1A2 (e.g., barbiturates, carbamazepine and rifampin) may decrease effect; cimetidine and antipsychotics may increase effect and toxicity; possible interaction with

Drug/Cost ^[a]	Dosage	Adverse Effects ^[b]	Drug Interactions
			antiarrhythmics (may lead to increased effect of either drug); may reduce antihypertensive effect of clonidine; may augment hypotensive effect of thiazides.
nortriptyline Aventyl \$60–90	Initial: ^[a] 25–50 mg/day PO Usual: 75–150 mg/day PO High: ^[a] 200 mg/day PO	Anticholinergic (dry mouth, blurred vision, constipation, urinary hesitancy, tachycardia, delirium), antihistaminergic (sedation, weight gain), orthostatic hypotension, lowered seizure threshold; sexual dysfunction.	Use with MAOIs , linezolid or methylene blue may lead to potentially fatal reaction initially presenting with tremor, agitation, hypomania, hyperthermia and/or hypertension. Inducers of CYP1A2 (e.g., barbiturates, carbamazepine and rifampin) may decrease effect; cimetidine and antipsychotics may increase effect and toxicity; possible interaction with antiarrhythmics (may lead to increased effect of either drug); may reduce antihypertensive effect of clonidine; may augment hypotensive effect of thiazides.
trimipramine generics \$30–60	Initial: ^[a] 25–50 mg/day PO Usual: 75–200 mg/day PO High: ^[a] 250–300 mg/day PO	Anticholinergic (dry mouth, blurred vision, constipation, urinary hesitancy, tachycardia, delirium), antihistaminergic (sedation, weight gain), orthostatic hypotension, lowered seizure	Use with MAOIs , linezolid or methylene blue may lead to potentially fatal reaction initially presenting with tremor, agitation, hypomania, hyperthermia and/or hypertension. Inducers of CYP1A2 (e.g., barbiturates, carbamazepine and rifampin) may decrease

Drug/Cost ^[a]	Dosage	Adverse Effects ^[b]	Drug Interactions
		threshold; sexual dysfunction.	effect; cimetidine and antipsychotics may increase effect and toxicity; possible interaction with antiarrhythmics (may lead to increased effect of either drug); may reduce antihypertensive effect of clonidine; may augment hypotensive effect of thiazides.
Drug Class: Natural Health Products			
S-adenosylmethionine (SAM-e) \$60–90	400–1600 mg/day PO, in 2–3 divided doses	Mild insomnia, constipation, dizziness, nausea, dry mouth, sweating; case reports of increased anxiety, mania or hypomania in patients with bipolar disorder.	Potential additive bleeding risk with drugs such as warfarin, ASA and other antiplatelet (including SSRIs) and anticoagulant agents. Avoid concurrent use with other serotonergic drugs such as antidepressants (possible serotonin syndrome).
St. John's wort (Hypericum perforatum) < \$30	Usual: 300 mg TID PO Range: 300–1800 mg/day PO, in 2–3 divided doses	Photosensitivity (rare), GI upset, dizziness, insomnia, restlessness, agitation; cases of mania or hypomania reported.	Avoid concurrent use with MAOIs , linezolid or methylene blue (possible hypertensive crisis) and SSRIs (possible serotonin syndrome). Inducer of many cytochrome P450 isoenzymes (1A2, 2C9, 2C19, 3A4) and P-glycoprotein; may decrease serum levels of many drugs including cyclosporine, digoxin, indinavir, oral contraceptives, theophylline and warfarin.

[a] Cost of 30-day supply for mean usual dose, unless otherwise specified; includes drug cost only.

[b] See also [Table 4](#) for relative incidence and management of adverse effects.

[c] Lower initial doses are indicated in patients who have previously experienced side effects or who are taking multiple medications; this often applies to older patients.

[d] Higher doses often exceed maximum recommended doses in manufacturers' product monographs and are usually associated with increased risk of adverse effects. These doses should be used with caution in appropriately selected patients.

[e] Esketamine is indicated for treatment-resistant depression, in combination with an SSRI or SNRI. It is available only through a controlled distribution program (JANSSEN JOURNEY). A trained health-care professional is required to supervise each dose and monitor patient clinical status and BP for at least 2 hours afterward.

[f] For SSRIs, the upper end of the "initial dose" may be the "usual dose," e.g., fluoxetine 20 mg or sertraline 50 mg; otherwise, increments are every 5–7 days.

 Dosage adjustment may be required in renal impairment; see [Dosage Adjustment in Renal Impairment](#).

Abbreviations: ASA = acetylsalicylic acid; BP = blood pressure; CNS = central nervous system; CYP = cytochrome P450; EPS = extrapyramidal side effects; GI = gastrointestinal; MAOI = monoamine oxidase inhibitor; NSAID = nonsteroidal anti-inflammatory drug; SIADH = syndrome of inappropriate antidiuretic hormone; SNRI = serotonin-norepinephrine reuptake inhibitor; SSRI = selective serotonin reuptake inhibitor

Suggested Readings

[Lam RW, Kennedy SH, Adams C et al. Canadian Network for Mood and Anxiety Treatments \(CANMAT\). 2023 update on clinical guidelines for management of major depressive disorder in adults. *Can J Psychiatry* 2024;69\(9\):641-87.](#)

[Pillinger T, Arumuham A, McCutcheon RA et al. The effects of antidepressants on cardiometabolic and other physiological parameters: a systematic review and network meta-analysis. *Lancet* 2025;406\(10515\):2063-77.](#)

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[Sobieraj DM, Martinez BK, Hernandez AV et al. Adverse effects of pharmacologic treatments of major depression in older adults. *J Am Geriatr Soc* 2019;67\(8\):1571-81.](#)

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